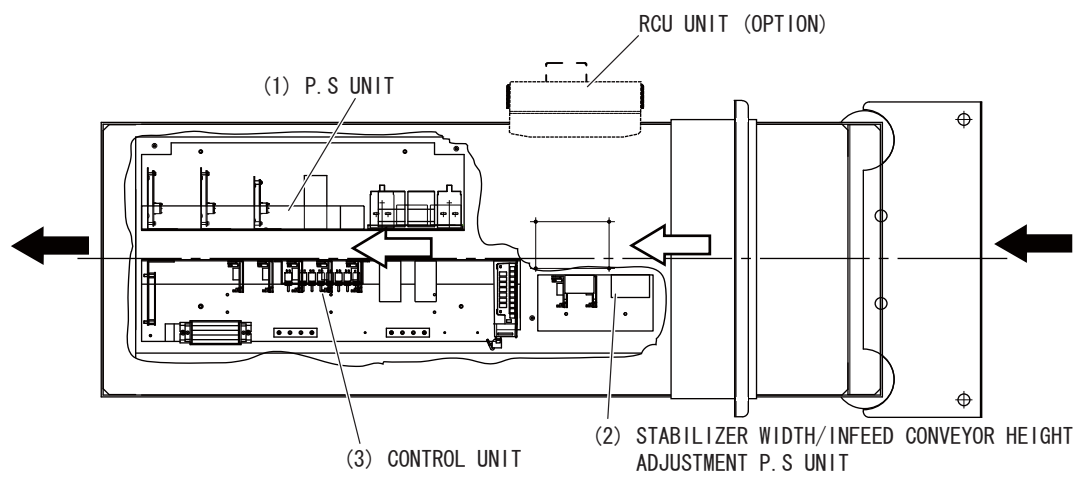
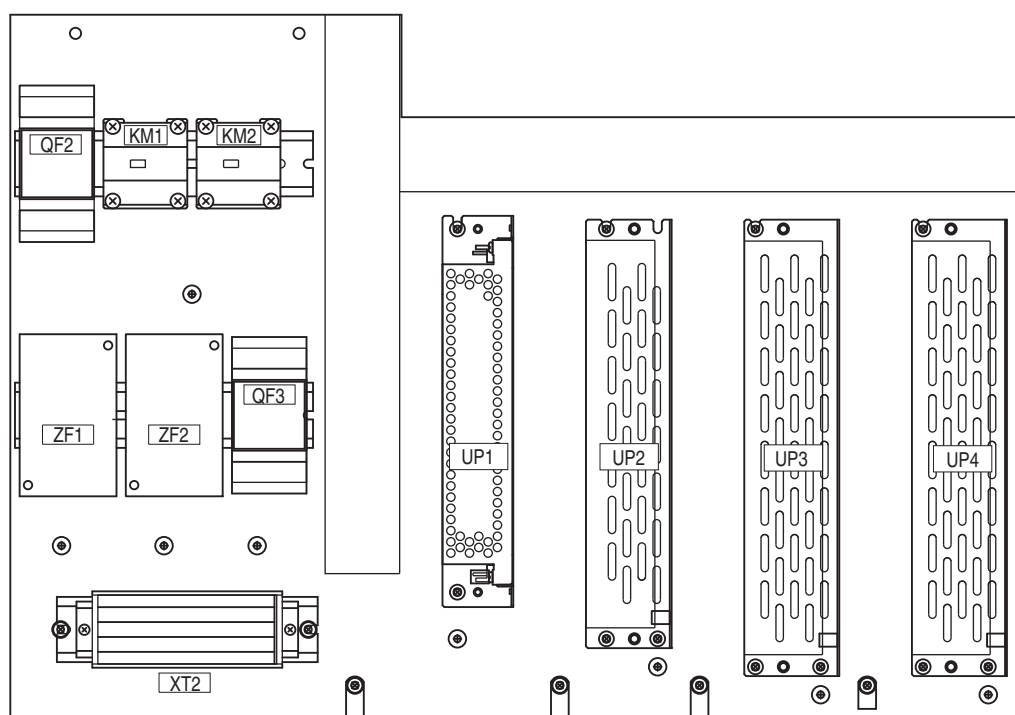


3 ELECTRICAL CONTROL UNIT

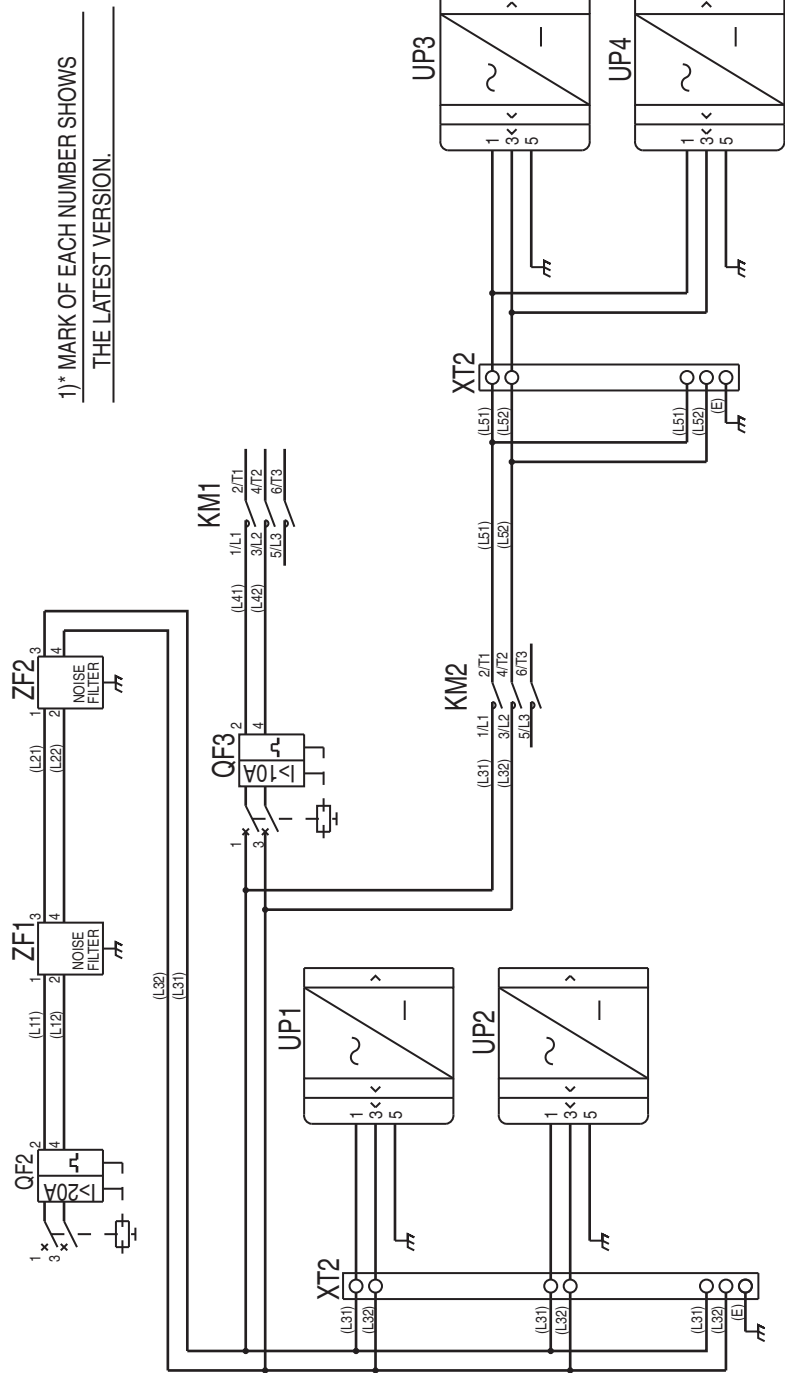
3.1 Locations of Electrical Devices



3.1.1 P. S UNIT



POWER SUPPLY UNIT

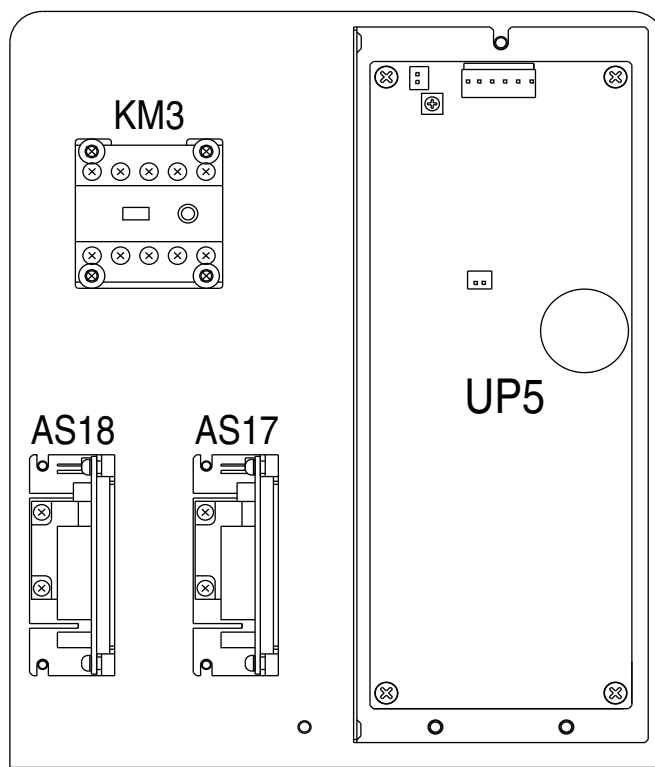


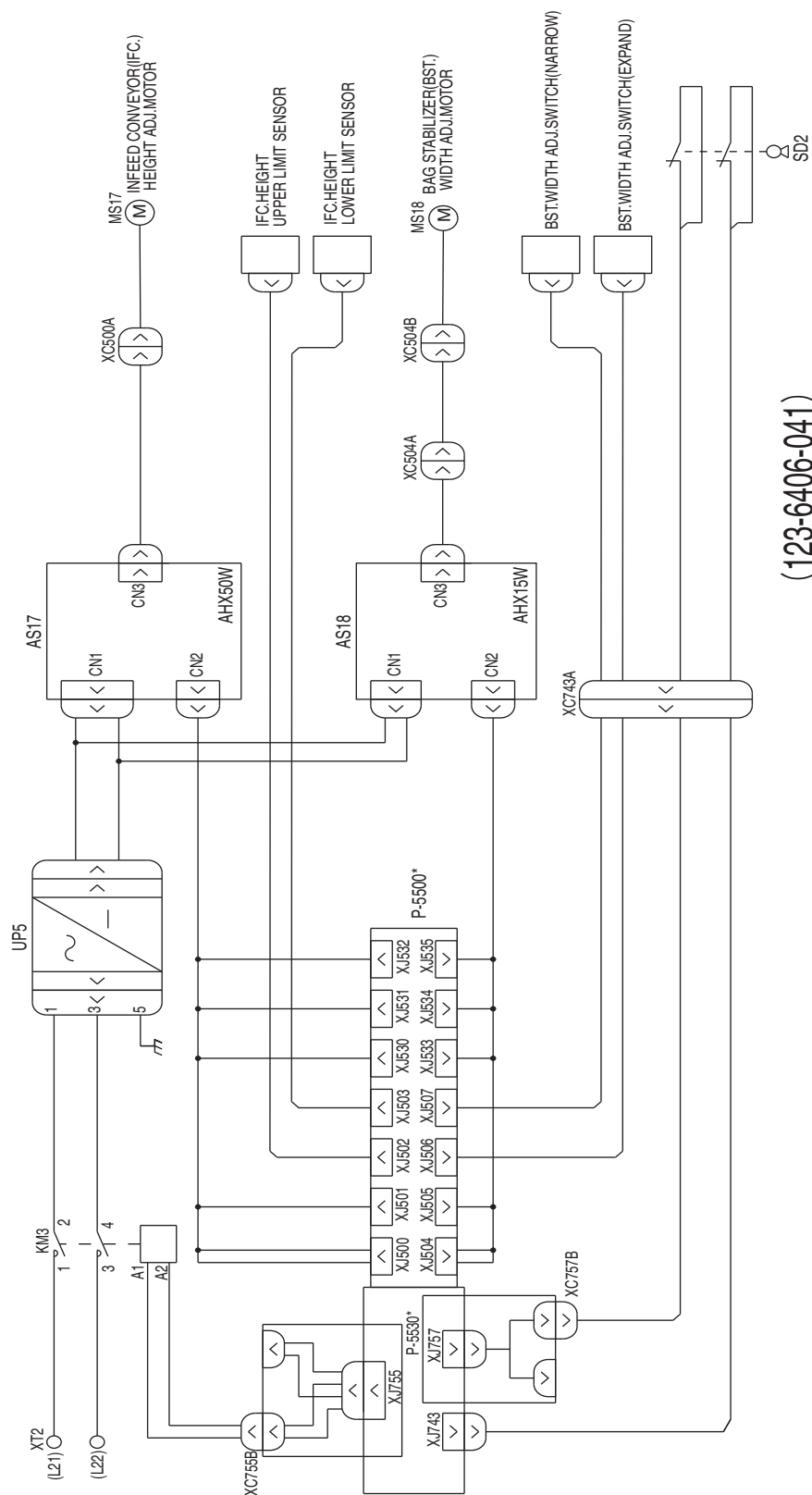
1)* MARK OF EACH NUMBER SHOWS
THE LATEST VERSION.

SYMBOL	DESCRIPTION	TYPE	MANUFAC.
KM1,2	MAGNETIC CONTACTOR	SD-Q11 DC24V	MITSUBISHI
UP1	POWER SUPPLY	LDC60F-2-SN	COSEL
UP2	POWER SUPPLY	ZWS150PAF-24/JL	DENSEI LAMBDA
UP3	POWER SUPPLY	ZWS240PAF-24/JL	DENSEI LAMBDA
UP4	POWER SUPPLY	ZWS240PAF-48/JL	DENSEI LAMBDA
ZF1,2	NOISE FILTER	NAP-20-472-D	COSEL
QF2	CIRCUIT BREAKER	NF30-FAU20A	MITSUBISHI
QF3	CIRCUIT BREAKER	NF30-FAU10A	MITSUBISHI
XT2	TERMINAL BLOCK	TPUJ2-20-12P	TOGI

POWER SUPPLY UNIT
BLOCK DIAGRAM:: (123-1792-042)

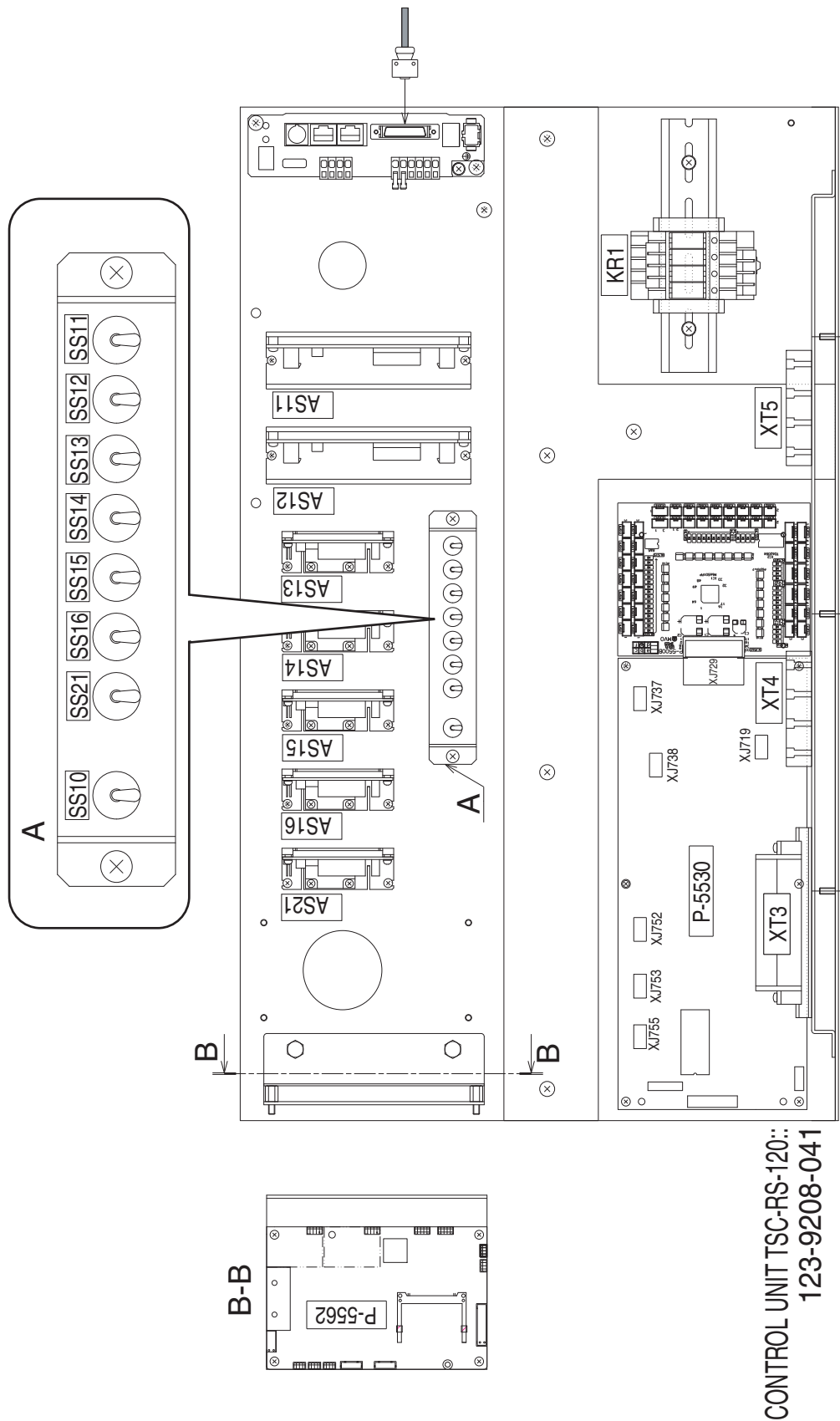
3.1.2 ADJUSTMENT P.S UNIT

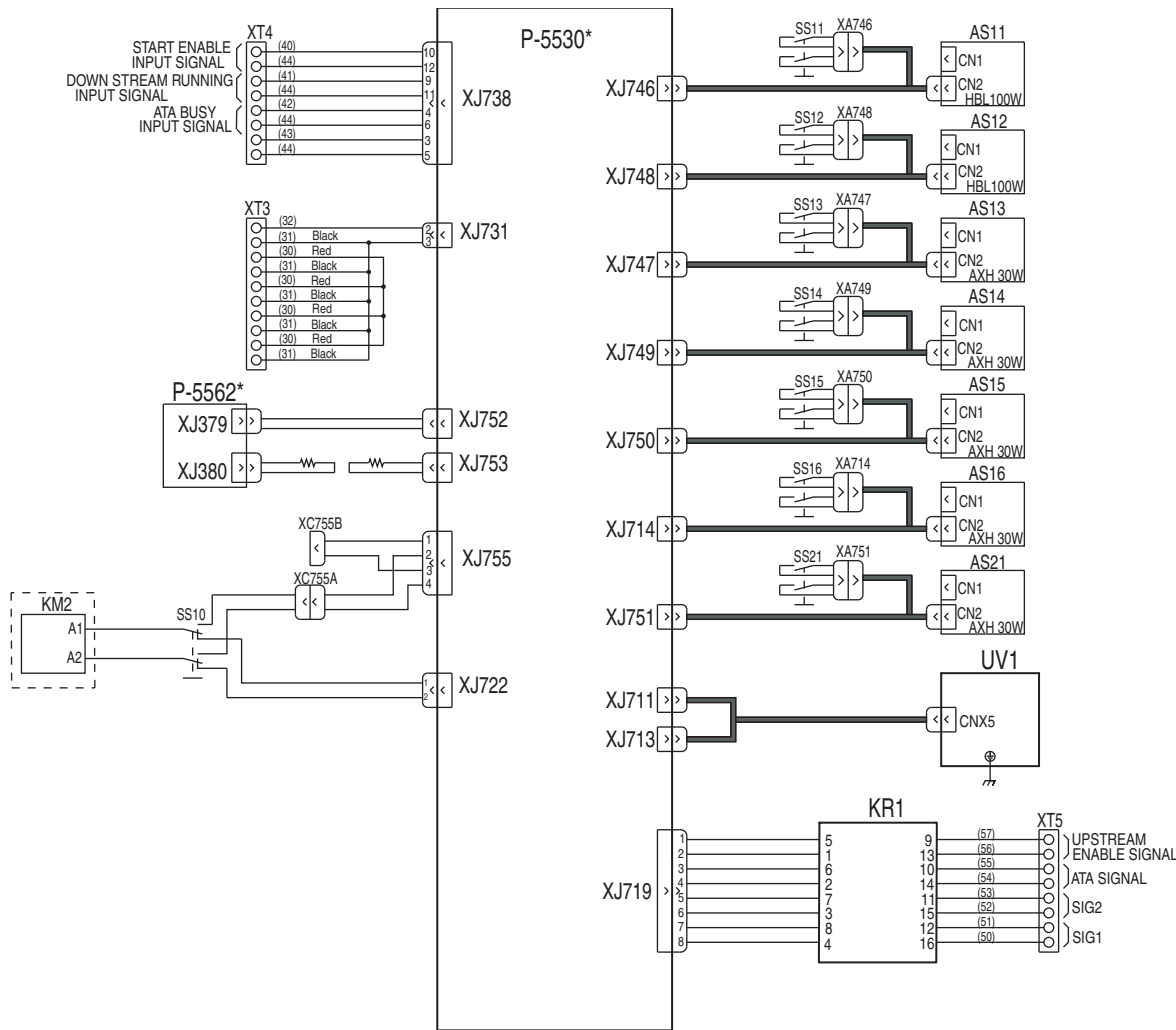




(123-6406-041)

3.1.3 CONTROL UNIT

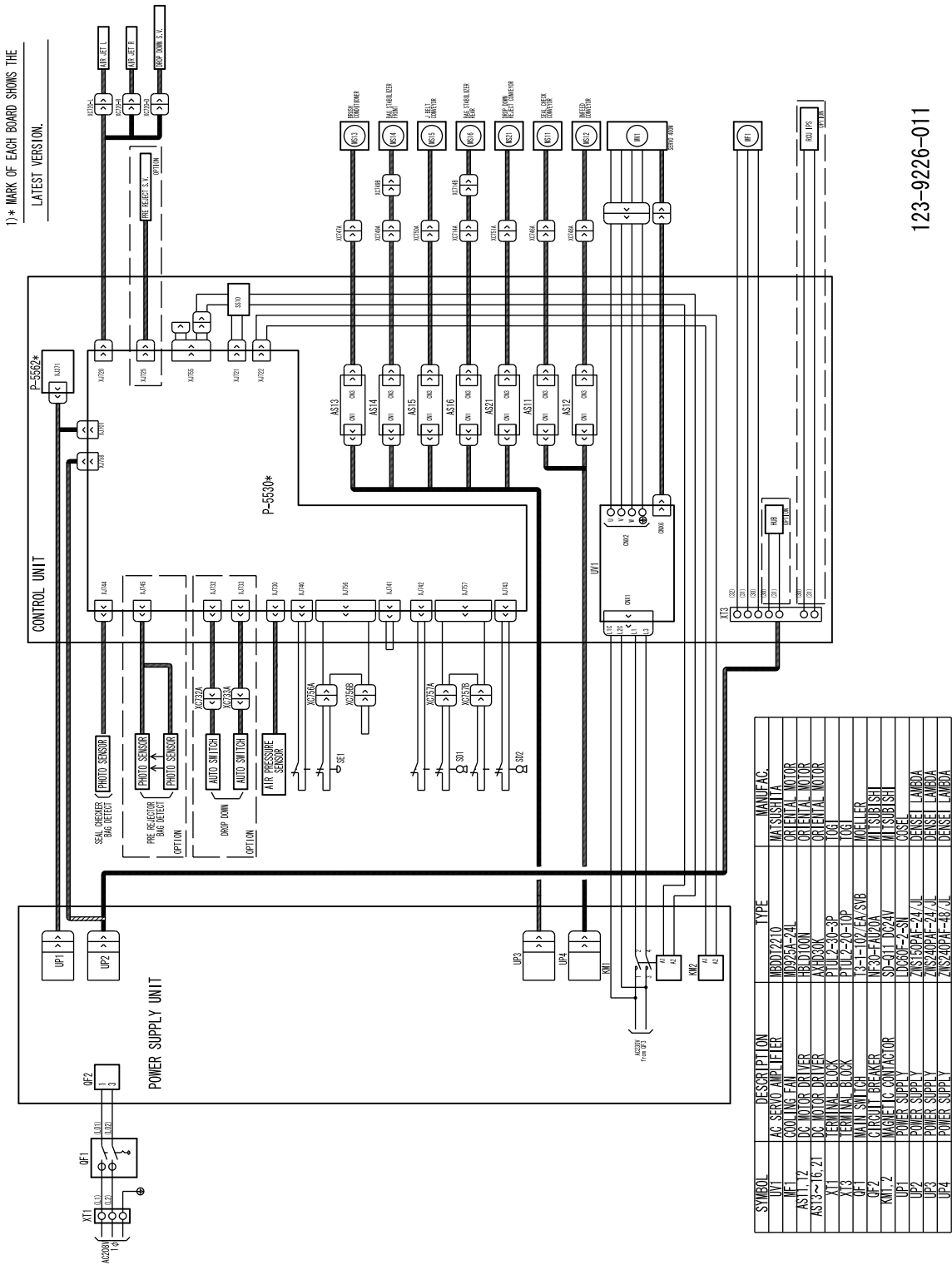




SYMBOL	DESCRIPTION	TYPE	MANUFAC.
UV1	AC SERVO AMPLIFIER	MADDT1207	Panasonic
AS11,12	DC MOTOR DRIVER	HBLD100N	ORIENTAL MOTOR
AS13-16,21	DC MOTOR DRIVER	AXHD30K	ORIENTAL MOTOR
SS10-16,21	MANUAL RUN SWITCH	M-2022	NKK
KR1	TERMINAL RELAY	G6B-4BND DC24V	OMRON
XT3	TERMINAL BLOCK	PTUL2-20-10P	TOGI
XT4,5	TERMINAL BLOCK	MK3/8KRG	

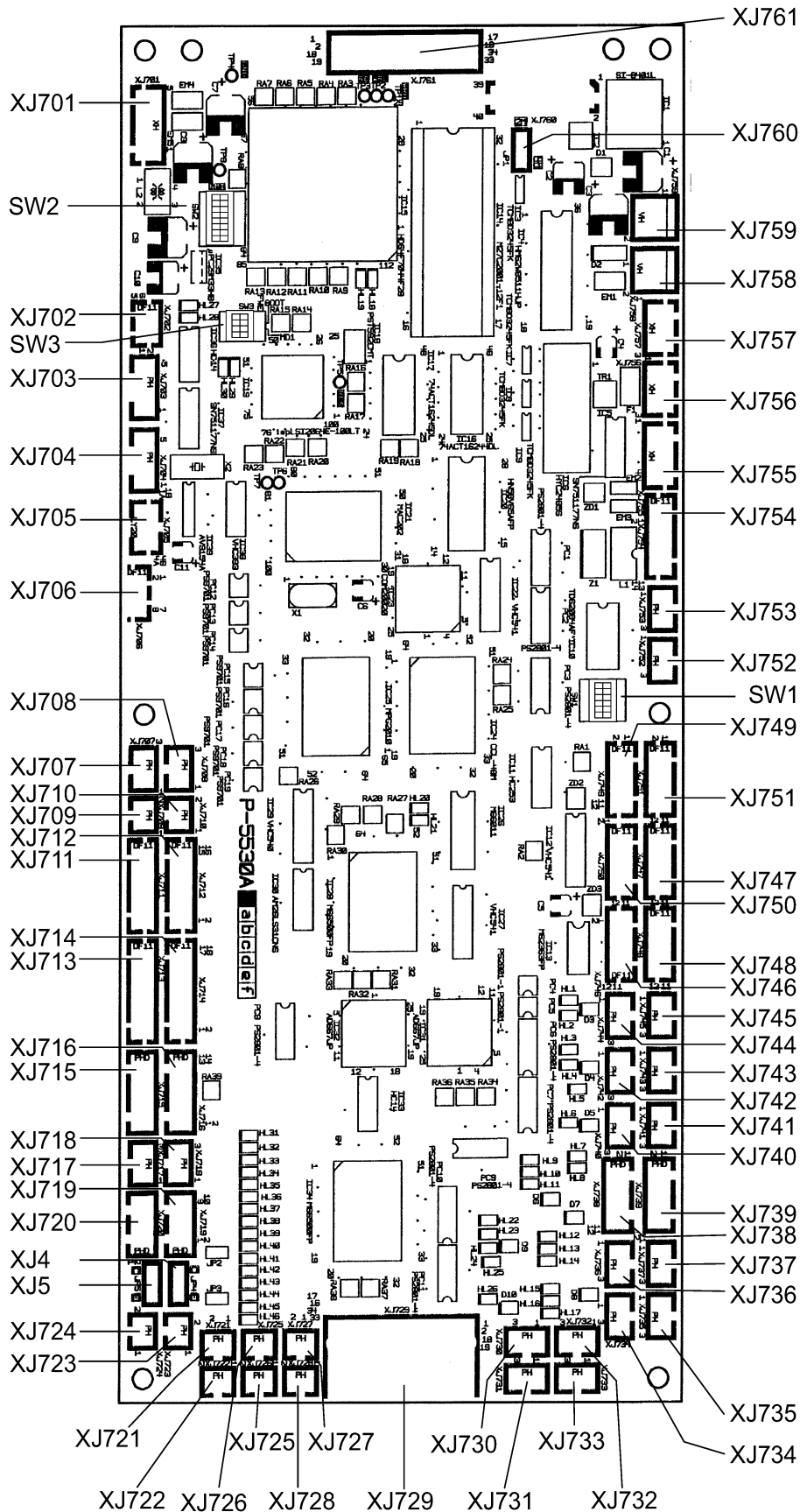
(123-9213-031)

3.1.4 TOTAL DIAGRAM



SYMBOL	DESCRIPTION	TYPE	MANUFACT.
UP1	AC SERVO AMPLIFIER	MD072210	MITSUBISHI
UP2	COOLING FAN	MD072210	MITSUBISHI
UP3	DC MOTOR DRIVER	MD072210	MITSUBISHI
UP4	DC MOTOR DRIVER	MD072210	MITSUBISHI
XT1	TERMINAL BLOCK	PLU2-30-3P	UGI
XT2	TERMINAL BLOCK	PLU2-30-3P	UGI
XT3	TERMINAL BLOCK	PLU2-30-3P	UGI
XT4	TERMINAL BLOCK	PLU2-30-3P	UGI
XT5	TERMINAL BLOCK	PLU2-30-3P	UGI
XT6	TERMINAL BLOCK	PLU2-30-3P	UGI
XT7	TERMINAL BLOCK	PLU2-30-3P	UGI
XT8	TERMINAL BLOCK	PLU2-30-3P	UGI
XT9	TERMINAL BLOCK	PLU2-30-3P	UGI
XT10	TERMINAL BLOCK	PLU2-30-3P	UGI
XT11	TERMINAL BLOCK	PLU2-30-3P	UGI
XT12	TERMINAL BLOCK	PLU2-30-3P	UGI
XT13	TERMINAL BLOCK	PLU2-30-3P	UGI
XT14	TERMINAL BLOCK	PLU2-30-3P	UGI
XT15	TERMINAL BLOCK	PLU2-30-3P	UGI
XT16	TERMINAL BLOCK	PLU2-30-3P	UGI
XT17	TERMINAL BLOCK	PLU2-30-3P	UGI
XT18	TERMINAL BLOCK	PLU2-30-3P	UGI
XT19	TERMINAL BLOCK	PLU2-30-3P	UGI
XT20	TERMINAL BLOCK	PLU2-30-3P	UGI
XT21	TERMINAL BLOCK	PLU2-30-3P	UGI
XT22	TERMINAL BLOCK	PLU2-30-3P	UGI
XT23	TERMINAL BLOCK	PLU2-30-3P	UGI
XT24	TERMINAL BLOCK	PLU2-30-3P	UGI
XT25	TERMINAL BLOCK	PLU2-30-3P	UGI
XT26	TERMINAL BLOCK	PLU2-30-3P	UGI
XT27	TERMINAL BLOCK	PLU2-30-3P	UGI
XT28	TERMINAL BLOCK	PLU2-30-3P	UGI
XT29	TERMINAL BLOCK	PLU2-30-3P	UGI
XT30	TERMINAL BLOCK	PLU2-30-3P	UGI
XT31	TERMINAL BLOCK	PLU2-30-3P	UGI
XT32	TERMINAL BLOCK	PLU2-30-3P	UGI
XT33	TERMINAL BLOCK	PLU2-30-3P	UGI
XT34	TERMINAL BLOCK	PLU2-30-3P	UGI
XT35	TERMINAL BLOCK	PLU2-30-3P	UGI
XT36	TERMINAL BLOCK	PLU2-30-3P	UGI
XT37	TERMINAL BLOCK	PLU2-30-3P	UGI
XT38	TERMINAL BLOCK	PLU2-30-3P	UGI
XT39	TERMINAL BLOCK	PLU2-30-3P	UGI
XT40	TERMINAL BLOCK	PLU2-30-3P	UGI
XT41	TERMINAL BLOCK	PLU2-30-3P	UGI
XT42	TERMINAL BLOCK	PLU2-30-3P	UGI
XT43	TERMINAL BLOCK	PLU2-30-3P	UGI
XT44	TERMINAL BLOCK	PLU2-30-3P	UGI
XT45	TERMINAL BLOCK	PLU2-30-3P	UGI
XT46	TERMINAL BLOCK	PLU2-30-3P	UGI
XT47	TERMINAL BLOCK	PLU2-30-3P	UGI
XT48	TERMINAL BLOCK	PLU2-30-3P	UGI
XT49	TERMINAL BLOCK	PLU2-30-3P	UGI
XT50	TERMINAL BLOCK	PLU2-30-3P	UGI
XT51	TERMINAL BLOCK	PLU2-30-3P	UGI
XT52	TERMINAL BLOCK	PLU2-30-3P	UGI
XT53	TERMINAL BLOCK	PLU2-30-3P	UGI
XT54	TERMINAL BLOCK	PLU2-30-3P	UGI
XT55	TERMINAL BLOCK	PLU2-30-3P	UGI
XT56	TERMINAL BLOCK	PLU2-30-3P	UGI
XT57	TERMINAL BLOCK	PLU2-30-3P	UGI
XT58	TERMINAL BLOCK	PLU2-30-3P	UGI
XT59	TERMINAL BLOCK	PLU2-30-3P	UGI
XT60	TERMINAL BLOCK	PLU2-30-3P	UGI
XT61	TERMINAL BLOCK	PLU2-30-3P	UGI
XT62	TERMINAL BLOCK	PLU2-30-3P	UGI
XT63	TERMINAL BLOCK	PLU2-30-3P	UGI
XT64	TERMINAL BLOCK	PLU2-30-3P	UGI
XT65	TERMINAL BLOCK	PLU2-30-3P	UGI
XT66	TERMINAL BLOCK	PLU2-30-3P	UGI
XT67	TERMINAL BLOCK	PLU2-30-3P	UGI
XT68	TERMINAL BLOCK	PLU2-30-3P	UGI
XT69	TERMINAL BLOCK	PLU2-30-3P	UGI
XT70	TERMINAL BLOCK	PLU2-30-3P	UGI
XT71	TERMINAL BLOCK	PLU2-30-3P	UGI
XT72	TERMINAL BLOCK	PLU2-30-3P	UGI
XT73	TERMINAL BLOCK	PLU2-30-3P	UGI
XT74	TERMINAL BLOCK	PLU2-30-3P	UGI
XT75	TERMINAL BLOCK	PLU2-30-3P	UGI
XT76	TERMINAL BLOCK	PLU2-30-3P	UGI
XT77	TERMINAL BLOCK	PLU2-30-3P	UGI
XT78	TERMINAL BLOCK	PLU2-30-3P	UGI
XT79	TERMINAL BLOCK	PLU2-30-3P	UGI
XT80	TERMINAL BLOCK	PLU2-30-3P	UGI
XT81	TERMINAL BLOCK	PLU2-30-3P	UGI
XT82	TERMINAL BLOCK	PLU2-30-3P	UGI
XT83	TERMINAL BLOCK	PLU2-30-3P	UGI
XT84	TERMINAL BLOCK	PLU2-30-3P	UGI
XT85	TERMINAL BLOCK	PLU2-30-3P	UGI
XT86	TERMINAL BLOCK	PLU2-30-3P	UGI
XT87	TERMINAL BLOCK	PLU2-30-3P	UGI
XT88	TERMINAL BLOCK	PLU2-30-3P	UGI
XT89	TERMINAL BLOCK	PLU2-30-3P	UGI
XT90	TERMINAL BLOCK	PLU2-30-3P	UGI
XT91	TERMINAL BLOCK	PLU2-30-3P	UGI
XT92	TERMINAL BLOCK	PLU2-30-3P	UGI
XT93	TERMINAL BLOCK	PLU2-30-3P	UGI
XT94	TERMINAL BLOCK	PLU2-30-3P	UGI
XT95	TERMINAL BLOCK	PLU2-30-3P	UGI
XT96	TERMINAL BLOCK	PLU2-30-3P	UGI
XT97	TERMINAL BLOCK	PLU2-30-3P	UGI
XT98	TERMINAL BLOCK	PLU2-30-3P	UGI
XT99	TERMINAL BLOCK	PLU2-30-3P	UGI
XT100	TERMINAL BLOCK	PLU2-30-3P	UGI

3.2 SSC CONTROL BOARD (P-5530*)



Function of LED for monitor

LED No.	Function / Description	LED display
HL1	PHOTO-1 Input monitor	Lights up during short-circuit of pin2 and pin3.
HL2	PHOTO-0 Input monitor	Lights up during short-circuit of pin2 and pin3.
HL3	EMG-3 Input monitor	Lights up during short-circuit of pin2 and pin3.
HL4	EMG-2 Input monitor	Lights up during short-circuit of pin2 and pin3.
HL5	EMG-1 Input monitor	Lights up during short-circuit of pin2 and pin3.
HL6	EMG-0 Input monitor	Lights up during short-circuit of pin2 and pin3.
HL7	GPI-15 Input monitor	GPI-15 : Lights up when it is connected to the PGND
HL8	GPI-14 Input monitor	GPI-14: Lights up when it is connected to the PGND
HL9	GPI-13 Input monitor	GPI-13: Lights up when it is connected to the PGND
HL10	GPI-12 Input monitor	GPI-12: Lights up when it is connected to the PGND
HL11	GPI-11 Input monitor	GPI-11 : Lights up when it is connected to the PGND
HL12	GPI-10 Input monitor	GPI-10 : Lights up when it is connected to the PGND
HL13	GPI-9 Input monitor	GPI-9 : Lights up when it is connected to the PGND
HL14	GPI-8 Input monitor	GPI-8 : Lights up when it is connected to the PGND
HL15	GPI-7 Input monitor	Lights up during short-circuit of pin2 and pin3.
HL16	GPI-6 Input monitor	Lights up during short-circuit of pin2 and pin3.
HL17	GPI-5 Input monitor	Lights up during short-circuit of pin2 and pin3.
HL18	Life LED	Flashes at 500msec interval in normal SSC operation
HL19	Copy LED	Lights up while copying software data to SRAM in the SSC when the SSC software (EPROM) is exchanged.
HL20	Seal check task monitor	Lights up in the seal check task operation. The dynamic check : from the time the product head passes the intake line sensor to the time it reaches the end of the check conveyor. The static check : from the time the product head passes the intake line sensor to the time the check conveyor restart after the pause.
HL21	(1) Seal check task monitor	Lights up in the seal check operation. The dynamic check : from the time after passing “the window start time” (90% of the product head is in the check head) to the time after passing “the window end time” (10% of the product head is out of the check head). The static check : from the time the product head passes the intake line sensor to the time the check conveyor restart after the pause.
	(2) Check head monitor	Lights up only once at the establishment of communication with DMU and SSC immediately after start-up TSC-RS.
HL22	GPI-1 Input monitor	Lights up during short-circuit of pin2 and pin3.
HL23	GPI-0 Input monitor	Lights up during short-circuit of pin2 and pin3.
HL24	GPI-2 Input monitor	Lights up during short-circuit of pin2 and pin3.

LED No.	Function / Description	LED display
HL25	GPI-3 Input monitor	Lights up during short-circuit of pin2 and pin3.
HL26	GPI-4 Input monitor	Lights up during short-circuit of pin2 and pin3.
HL27	ARCNET monitor	Lights up during data transmission.
HL28	ARCNET monitor	Lights up during data transmission.
HL29	Not used for TSC-RS	
HL30	Not used for TSC-RS	
HL31	GPO-15 Input monitor	Lights up at the output GPO-15
HL32	GPO-14 Input monitor	Lights up at the output GPO-14
HL33	GPO-13 Input monitor	Lights up at the output GPO-13
HL34	GPO-12 Input monitor	Lights up at the output GPO-12
HL35	GPO-11 Input monitor	Lights up at the output GPO-11
HL36	GPO-10 Input monitor	Lights up at the output GPO-10
HL37	GPO-9 Input monitor	Lights up at the output GPO-9
HL38	GPO-8 Input monitor	Lights up at the output GPO-8
HL39	GPO-7 Input monitor	Lights up at the output GPO-7
HL40	GPO-6 Input monitor	Lights up at the output GPO-6
HL41	GPO-5 Input monitor	Lights up at the output GPO-5
HL42	GPO-4 Input monitor	Lights up at the output GPO-4
HL43	GPO-3 Input monitor	Lights up at the output GPO-3
HL44	GPO-2 Input monitor	Lights up at the output GPO-2
HL45	GPO-1 Input monitor	Lights up at the output GPO-1
HL46	GPO-0 Input monitor	Lights up at the output GPO-0

Category of Connectors

Connector No.		Category
XJ701	(5P)	DC5V.+DC15V.-15V(initial) power receiving
XJ706	(8P)	RS communication CHO
XJ711	(16P)	Servo motor For the Pulse and I/O
XJ713	(18P)	Servo motor DA (-12V~+12V) • I/O
XJ714	(18P)	Servo motor DA (-12V~+12V) • I/O
XJ719	(10P)	General-purpose Input 8~11
XJ720	(10P)	General-purpose Input 12~15
XJ721	(2P)	General-purpose Input 0
XJ722	(2P)	General-purpose Input 1
XJ723	(2P)	General-purpose Input 2
XJ724	(2P)	General-purpose Input 3
XJ725	(2P)	General-purpose Input 4
XJ726	(2P)	General-purpose Input 5
XJ727	(2P)	General-purpose Input 6
XJ728	(2P)	General-purpose Input 7
XJ730	(3P)	General-purpose Input 0
XJ731	(3P)	General-purpose Input 1
XJ732	(3P)	General-purpose Input 2
XJ733	(3P)	General-purpose Input 3
XJ734	(3P)	General-purpose Input 4
XJ735	(3P)	General-purpose Input 5
XJ736	(3P)	General-purpose Input 6
XJ737	(3P)	General-purpose Input 7
XJ738	(12P)	General-purpose Input 8~11
XJ739	(12P)	General-purpose Input 12~15
XJ740	(3P)	Emergency stop class input 0
XJ741	(3P)	Emergency stop class input 1
XJ742	(3P)	Emergency stop class input 2
XJ743	(3P)	Emergency stop class input 3
XJ744	(3P)	Interrupt Input 0
XJ745	(3P)	Interrupt Input 1
XJ746	(12P)	DC motor
XJ747	(12P)	DC motor
XJ748	(12P)	DC motor
XJ749	(12P)	DC motor
XJ750	(12P)	DC motor
XJ751	(12P)	DC motor
XJ752	(3P)	ARCNET communication
XJ753	(3P)	ARCNET communication
XJ755	(4P)	Power range system 24V external distribution
XJ756	(3P)	Power range system 24V P-5530 internal distribution
XJ757	(3P)	Power range system 24V P-5530 internal distribution
XJ758	(2P)	24V power receiving / power distribution
XJ759	(2P)	24V power receiving / power distribution

Connector function in details

(1) XJ701 : 5V • +15V • -15V (initial) power receiving (XH connector 5P)

Connector No.	Terminal No.	Signal
XJ701 DC5V • +DC15V, -15V (initial) power receiving	1	VCC (DC5V)
	2	GND
	3	+A15V
	4	AGND
	5	-A15V

(2) XJ706 : RS323 communication CH0 (DF11 connector 8P)

Connector No.	Terminal No.	Signal
XJ706 RS232 communication CHO	1	VCC
	2	T DO x
	3	x O D
	4	GND
	5	/RTS
	6	/CTS
	7	/DSR
	8	/DTR

(3) XJ711 : Servo motor For the Pulse and I/O (DF11 connector 16P)

- Design for MITSUBISHI MR-J2S connector CN1A on the side of driver

Connector No.	Terminal No.	Signal
XJ711 Servo motor For the Pulse and I/O	1	LG (AGND)
	2	OPC
	3	NP
	4	NG
	5	PP
	6	PG
	7	LA
	8	LAR
	9	LB
	10	LBR
	11	CR/SP1
	12	INP/SA
	13	COM (24V)
	14	RD
	15	SG (PGND)
	16	SG (PGND)

(4) XJ713 : Servo motor For DA (-12V~+12V • -15V~+15V) and I/O (DF11 connector 18P)

- Design for MITSUBISHI MR-J2S connector CN1B on the side of driver TSC-RS-120: -12V~+12V used

Connector No.	Terminal No.	Signal
XJ713 Servo motor For DA (-12V~+12V • -15V~+15V) and I/O	1	LG (AGND)
	2	VC / VLA (-12~+12V)
	3	TLA / TC (-12~+12V)
	4	COM (24V)
	5	DO1
	6	RES
	7	SON
	8	EMG (EGND)
	9	TLC LC / V
	10	LSP
	11	LOP / SP2
	12	LSN
	13	PC / ST1 / RS2
	14	ALM
	15	TL / ST2 / RS1
	16	ZSP / MBR
	17	SG (PGND)
	18	SG (PGND)

(5) XJ714 : Servo motor For DA (-12V~ +12V) and I/O (DF11 connector 18P)

- Design for MITSUBISHI MR-J2S connector CN1B on the side of driver TSC-RS-120: -12V~ +12V used TSC-RS-120 uses this for DC brushless motor control after stabilizer using analog output and I/O.

Connector No.	Terminal No.	Signal
XJ714 Servo motor For DA (-12V~+12V) and I/O	1	LG (AGND)
	2	VC / VLA (-12~+12V)
	3	TLA / TC (-12~+12V)
	4	COM (24V)
	5	DO1
	6	RES
	7	SON
	8	EMG (EGND)
	9	TLC LC / V
	10	LSP
	11	LOP / SP2
	12	LSN
	13	PC / ST1 / RS2
	14	ALM
	15	TL / ST2 / RS1
	16	ZSP / MBR
	17	SG (PGND)
	18	SG (PGND)

(6) XJ719 : General-purpose output 8~11 (PHD connector 10P)

Connector No.	Terminal No.	Signal
XJ719 General-purpose output 8~11	1	+24V
	2	GPO-15
	3	+24V
	4	GPO-14
	5	+24V
	6	GPO-13
	7	+24V
	8	GPO-12
	9	PGND
	10	PGND

(7) XJ720 : General-purpose output 12~15 (PHD connector 10P)

Connector No.	Terminal No.	Signal
XJ720 General-purpose output 12~15	1	+24V
	2	GPO-15
	3	+24V
	4	GPO-14
	5	+24V
	6	GPO-13
	7	+24V
	8	GPO-12
	9	PGND
	10	PGND

(8) XJ721 : General-purpose output 0 (PH connector 2P)

Connector No.	Terminal No.	Signal
XJ721 General-purpose output 0	1	+24V
	2	GPO-0

(9) XJ722 : General-purpose output 1 (PH connector 2P)

Connector No.	Terminal No.	Signal
XJ722 General-purpose output 1	1	+24V
	2	GPO-1

(10) XJ723 : General-purpose output 2 (PH connector 2P)

Connector No.	Terminal No.	Signal
XJ723 General-purpose output 2	1	+24V
	2	GPO-2

(11) XJ724 : General-purpose output 3 (PH connector 2P)

Connector No.	Terminal No.	Signal
XJ724 General-purpose output 3	1	+24V
	2	GPO-3

(12) XJ725 : General-purpose output 4 (PH connector 2P)

Connector No.	Terminal No.	Signal
XJ725 General-purpose output 4	1	+24V
	2	GPO-4

(13) XJ726 : General-purpose output 5 (PH connector 2P)

Connector No.	Terminal No.	Signal
XJ726 General-purpose output 5	1	+24V
	2	GPO-5

(14) XJ727 : General-purpose output 6 (PH connector 2P)

Connector No.	Terminal No.	Signal
XJ727 General-purpose output 6	1	+24V
	2	GPO-6

(15) XJ728 : General-purpose output 7 (PH connector 2P)

Connector No.	Terminal No.	Signal
XJ728 General-purpose output 7	1	+24V
	2	GPO-7

(16) XJ730 : General-purpose input 0 (PH connector 3P)

Connector No.	Terminal No.	Signal
XJ730 General-purpose input 0	1	+24V
	2	GPI-0
	3	PGND

(17) XJ731 : General-purpose input 1 (PH connector 3P)

Connector No.	Terminal No.	Signal
XJ731 General-purpose input 1	1	+24V
	2	GPI-1
	3	PGND

(18) XJ732 : General-purpose input 2 (PH connector 3P)

Connector No.	Terminal No.	Signal
XJ732 General-purpose input 2	1	+24V
	2	GPI-2
	3	PGND

(19) XJ733 : General-purpose input 3 (PH connector 3P)

Connector No.	Terminal No.	Signal
XJ733 General-purpose input 3	1	+24V
	2	GPI-3
	3	PGND

(20) XJ734 : General-purpose input 4 (PH connector 3P)

Connector No.	Terminal No.	Signal
XJ734 General-purpose input 4	1	+24V
	2	GPI-4
	3	PGND

(21) XJ735 : General-purpose input 5 (PH connector 3P)

Connector No.	Terminal No.	Signal
XJ735 General-purpose input 5	1	+24V
	2	GPI-5
	3	PGND

(22) XJ736 : General-purpose input 6 (PH connector 3P)

Connector No.	Terminal No.	Signal
XJ736 General-purpose input 6	1	+24V
	2	GPI-6
	3	PGND

(23) XJ737 : General-purpose input 7 (PH connector 3P)

Connector No.	Terminal No.	Signal
XJ737 General-purpose input 7	1	+24V
	2	GPI-7
	3	PGND

(24) XJ738 : General-purpose input 8~11 (PHD connector 12P)

Connector No.	Terminal No.	Signal
XJ738 General-purpose input 8~11	1	+24V
	2	+24V
	3	PGI-11
	4	PGI-10
	5	PGND
	6	PGND
	7	+24V
	8	+24V
	9	PGI-9
	10	PGI-8
	11	PGND
	12	PGND

(25) XJ739 : General-purpose input 12~15 (PHD connector 12P)

Connector No.	Terminal No.	Signal
XJ739 General-purpose input 12~15	1	+24V
	2	+24V
	3	PGI-15
	4	PGI-14
	5	PGND
	6	PGND
	7	+24V
	8	+24V
	9	PGI-13
	10	PGI-12
	11	PGND
	12	PGND

(26) XJ740 : Emergency stop class input 0 (PH connector 3P)

Connector No.	Terminal No.	Signal
XJ740 Emergency stop class input 0	1	+24V
	2	EMG-0
	3	PGND

(27) XJ741 : Emergency stop class input 1 (PH connector 3P)

Connector No.	Terminal No.	Signal
XJ741 Emergency stop class input 1	1	+24V
	2	EMG-1
	3	PGND

(28) XJ742 : Emergency stop class input 2 (PH connector 3P)

Connector No.	Terminal No.	Signal
XJ742 Emergency stop class input 2	1	+24V
	2	EMG-2
	3	PGND

(29) XJ743 : Emergency stop class input 3 (PH connector 3P)

Connector No.	Terminal No.	Signal
XJ743 Emergency stop class input 3	1	+24V
	2	EMG-3
	3	PGND

(30) XJ744 : Interrupt input 0 (PH connector 3P)

- Used for the check line sensor.
The sensor is installed between the intake conveyor and the check conveyor.

Connector No.	Terminal No.	Signal
XJ744 Interrupt input 0	1	+24V
	2	PHOTO-0
	3	PGND

(31) XJ745 : Interrupt input 1 (PH connector 3P)

- Used as line sensor input for the rejector.
The sensor is installed between the J-belt conveyor and the intake conveyor.

Connector No.	Terminal No.	Signal
XJ745 Interrupt input 1	1	+24V
	2	PHOTO-1
	3	PGND

(32) XJ746 : DC motor (DF11 connector 12P) For DC motor

Connector No.	Terminal No.	Signal
XJ746 DC motor	1	+5V
	2	EGND
	3	RUN0-OUT
	4	PGND-SW0
	5	RESET-OUT
	6	PGND
	7	VRM0-OUT (0V~5V)
	8	PGND
	9	SPEED0-IN
	10	PGND
	11	ALARM0-IN
	12	PGND

(33) XJ747 : DC motor (DF11 connector 12P) For DC motor

Connector No.	Terminal No.	Signal
XJ747 DC motor	1	+5V
	2	EGND
	3	RUN5-OUT
	4	PGND-SW5
	5	RESET-OUT
	6	PGND
	7	VRM5-OUT (0V~5V)
	8	PGND
	9	SPEED5-IN
	10	PGND
	11	ALARM5-IN
	12	PGND

(34) XJ748 : DC motor (DF11 connector 12P) For DC motor

Connector No.	Terminal No.	Signal
XJ748 DC motor	1	+5V
	2	EGND
	3	RUN2-OUT
	4	PGND-SW2
	5	RESET-OUT
	6	PGND
	7	VRM2-OUT (0V~5V)
	8	PGND
	9	SPEED2-IN
	10	PGND
	11	ALARM2-IN
	12	PGND

(35) XJ749 : DC motor (DF11 connector 12P) For DC motor

Connector No.	Terminal No.	Signal
XJ749 DC motor	1	+5V
	2	EGND
	3	RUN3-OUT
	4	PGND-SW3
	5	RESET-OUT
	6	PGND
	7	VRM3-OUT (0V~5V)
	8	PGND
	9	SPEED3-IN
	10	PGND
	11	ALARM3-IN
	12	PGND

(36) XJ750 : DC motor (DF11 connector 12P) For DC motor

Connector No.	Terminal No.	Signal
XJ750 DC motor	1	+5V
	2	EGND
	3	RUN4-OUT
	4	PGND-SW4
	5	RESET-OUT
	6	PGND
	7	VRM4-OUT (0V~5V)
	8	PGND
	9	SPEED4-IN
	10	PGND
	11	ALARM4-IN
	12	PGND

(37) XJ751 : DC motor (DF11 connector 12P) For DC motor

Connector No.	Terminal No.	Signal
XJ751 DC motor	1	+5V
	2	EGND
	3	RUN5-OUT
	4	PGND-SW5
	5	RESET-OUT
	6	PGND
	7	VRM5-OUT (0V~5V)
	8	PGND
	9	SPEED5-IN
	10	PGND
	11	ALARM5-IN
	12	PGND

(38) XJ752 : ARCNET Communication (PH connector 3P)

Connector No.	Terminal No.	Signal
XJ752 ARCNET Communication	1	ARCNET (D)
	2	ARCNET (/D)
	3	SGND

(39) XJ753 : ARCNET Communication (PH connector 3P)

Connector No.	Terminal No.	Signal
XJ753 ARCNET Communication	1	ARCNET (D)
	2	ARCNET (/D)
	3	SGND

(40) XJ755 : Power range system 24V external distribution (XH connector 4P)

Connector No.	Terminal No.	Signal
XJ755 Power range system 24V external distribution	1	+E24V
	2	+E24V
	3	PGND
	4	PGND

(41) XJ756 : Power range system 24V P-5530 internal distribution (PH connector 3P)

- 24V is supplied into the internal P-5530 by BOTH short-circuiting PIN1 and PIN2 of XJ756 AND short-circuiting PIN1 and PIN2 of XJ757

Connector No.	Terminal No.	Signal
XJ756 Power range system 24V P-5530 internal distribution	1	EMG-B1
	2	EMG-B2
	3	PGND

(42) XJ757 : Power range system 24V P-5530 internal distribution (PH connector 3P)

- 24V is supplied into XJ756 by short-circuiting PIN1 and PIN2 of XJ757

Connector No.	Terminal No.	Signal
XJ756 Power range system 24V P-5530 internal distribution	1	EMG-A1
	2	EMG-A2
	3	PGND

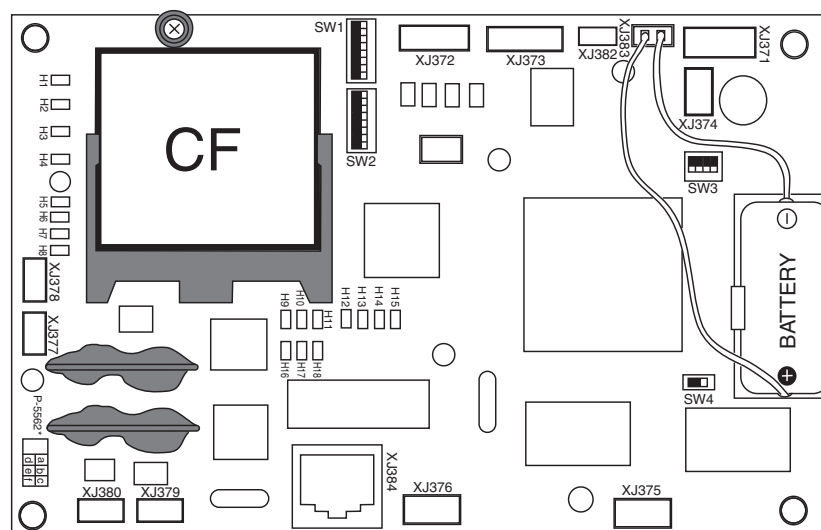
(43) XJ758 : 24V power receiving / power distribution (VH connector 2P)

Connector No.	Terminal No.	Signal
XJ758 DC+24V power receiving / power distribution	1	DC+24V
	2	PGND

(44) XJ759 : 24V power receiving / power distribution (VH connector 2P)

Connector No.	Terminal No.	Signal
XJ759 DC+24V power receiving / power distribution	1	DC+24V
	2	PGND

3.3 DMU BOARD (P-5562*)



Function of board

1. Data processing
2. Memory of Total and Parameter

Setting board DIP switches

(1) Setting SW1 / SW2 / SW3

DIP SW	Function	Default at shipment
SW 1 - 1 to SW 1 - 8	Not used	OFF
SW 2-1 to SW 2 - 2	Not used	OFF
SW 2-3	Set the MESLEC reconnecting function to ON.	ON
SW 2-4	ON: Use inspection head with stroke of 120 mm OFF: Use inspection head with stroke of 80 mm	ON
SW2-5	ON: Does not output FIF 0 at DMU application OFF: Outputs FIF 0 at DMU application	OFF
SW2-6	ON: Clear SRAM on memory initialization OFF: Does not clear SRM on memory initialization	OFF
SW2-7	ON: Set default IP address (192.168.0.31) OFF: Does not set default IP address	OFF
SW2-8	ON: Does not output message of boot loader OFF: Outputs message of boot loader	OFF
SW3-1	ON: Boot loader mode OFF: Application execution mode	OFF
SW3-2	Fix at OFF; do not turn ON.	OFF
SW3-3	ON: Flash memory write is inhibited. OFF: Flash memory write is enabled.	OFF
SW3-4	Fix at OFF; do not turn ON.	OFF

(4) Setting SW4 (Normally OFF)

Writing flash memory	OFF: Writing prohibited (available only with R23 mounted)
	ON: Writing available

Battery

Buttery in use: Lithium battery (Storage battery) CR17335SE-H SANYO

When replacing battery, connect new battery to XJ41. If an old battery is already connected, connect new battery to XJ40. Then remove old battery from another connector.



**If the battery is replaced with an incorrect one, it may result in a board malfunction. When replacing a battery, make sure that it is the same model or an equivalent.
Dispose of used batteries after replacement.**



Procedure for exchange of DMU board is as follows:

- (1) Turn ON DIP-SW2-3, DIP-SW2-4 (only Universal specification), DIP-SW2-6, DIP-SW2-7, DIP-SW3-3.
- (2) Insert the CF to card slot, then turn ON Power supply.
Installation will be started automatically.
- (3) Please wait until WEB-RCU is booted up.
- (4) (Shifting to TSC-RS-120 screen in case of ITPS system)
- (5) Turn OFF DIP-SW2-6, DIP-SW2-7, DIP-SW3-3. (With Universal specification, Keep only DIP-SW2-3, DIP-SW2-4 ON).
Turn ON Main power supply.
- (6) Shift to the screen [Memory initialization & Copy],
and set back to the setting in shipment from factory with pressing [load] button in the setting of Master memory.

Function of LED for monitor

LED	Functional description	LED display status
H1	PINT2	
H2	PINT3	
H3	PINT0	
H4	PINT1	
H5	Arcnet CH1 Reception signal	Normally, always ON. Abnormal when blinking in approx. every second.
H6	Arcnet CH1 Transmission signal	
H7	Arcnet CH0 Reception signal	
H8	Arcnet CH0 Transmission signal	
H12	Indicates the boot loader status by combination of 12 to 15.	LED lights ON/OFF in the order from 12 to 15. The boot loader causes error when all of four LEDs are blinking. Turning ON of all four LEDs starts the application.
H13		
H14		
H15		
H9	ETHER data reception	Normally blinking.
H10	ETHER link	Normally ON.
H11	ETHER all double communication	Normally OFF.
H16	ETHER link with 100MBPS	
H17	ETHER data transmission	Normally blinking.
H18	ETHER collision occurred	Normally OFF.

Connector classification

Connector No.		Item
XJ370	(6P)	JTAG for CPLD (IC8)
XJ371	(4P)	DMU power supply
XJ372	(10P)	PI input CH1
XJ373	(12P)	PO output CH1
XJ374	(8P)	JTAG for CPU
XJ375	(8P)	Serial interface CH0
XJ376	(8P)	Serial interface CH1
XJ377	(3P)	Arcnet CH1
XJ378	(3P)	Arcnet CH1
XJ379	(3P)	Arcnet CH0
XJ380	(3P)	Arcnet CH0
XJ381		Compact flash
XJ382	(2P)	Battery
XJ383	(2P)	Battery
XJ384		ETHERNET connector
XJ385	(20P)	Interface for DMU
XJ386	(10P)	Power supply for stack

Connector function in details

(1) XJ370: JTAG for CPLD (IC8)

Connector No.	Terminal No.	Signal
XJ370 JTAG for CPLD (IC8)	1	+3.3V
	2	TMS
	3	TDO
	4	TDI
	5	TCK
	6	GND

(2) XJ371: MDU power supply

Connector No.	Terminal No.	Signal
XJ371 DMU power supply	1	+5V C D
	2	+5V C D
	3	GND
	4	GND

(3) XJ372: PI input CH1

Connector No.	Terminal No.	Signal
XJ372 PI input CN1	1	+ ommon c
	2	PINT 0
	3	PINT 1
	4	PINT 2
	5	PINT 3
	6	PINT 4
	7	PINT 5
	8	PINT 6
	9	PINT 7
	10	N.C.

(4) XJ373: PO input CH1

Connector No.	Terminal No.	Signal
XJ373 PO output CH1	1	Data 0
	2	Data 1
	3	Data 2
	4	Data 3
	5	Data 4
	6	Data 5
	7	Data 6
	8	N.C.
	9	Data 7
	10	+ 5V
	11	GND
	12	N.C.

(5) XJ374: JTAG for CPU

Connector No.	Terminal No.	Signal
XJ374 JTAG for CPU	1	TCK
	2	TRST
	3	TDO
	4	GND
	5	TMS
	6	TDI
	7	RES
	8	GND

(6) XJ375: Serial interface CH0

Connector No.	Terminal No.	Signal
XJ375 Serial interface CH0	1	+5V
	2	TXD
	3	RXD
	4	GND
	5	RTS
	6	CTS
	7	DSR
	8	DTR

(7) XJ376: Serial interface CH1

Connector No.	Terminal No.	Signal
XJ376 Serial interface CH1	1	+5V
	2	TXD
	3	RXD
	4	GND
	5	RTS
	6	CTS
	7	DSR
	8	DTR

(8) XJ377/ XJ378: Arcnet CH1

Connector No.	Terminal No.	Signal
XJ377, XJ378 Arcnet CH1	1	Signal (+)
	2	Signal (-)
	3	F.G.

(9) XJ379/XJ380: Arcnet CH0

Connector No.	Terminal No.	Signal
XJ379, XJ380 Arcnet CH0	1	Signal (+)
	2	Signal (-)
	3	F.G.

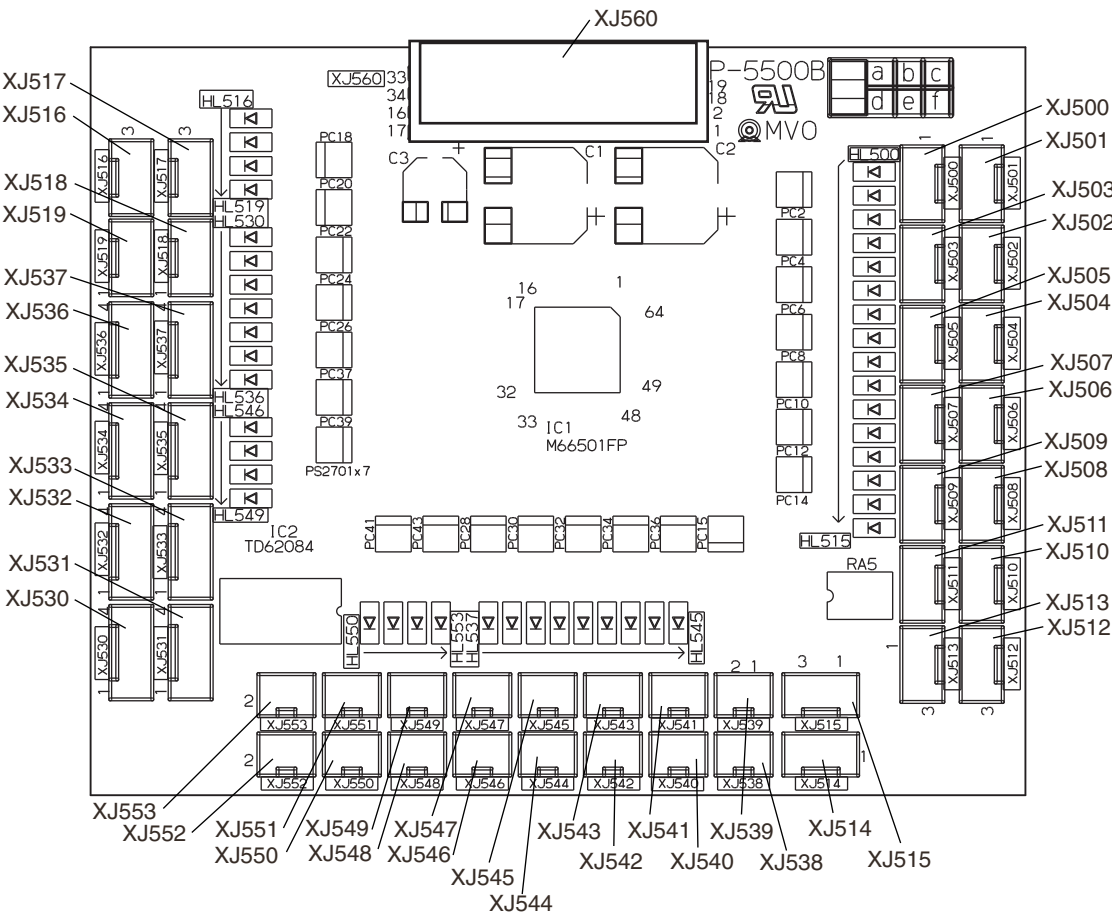
(10) XJ385: Interface for DMU

Connector No.	Terminal No.	Signal
XJ385 Interface for DMU	1	Data 0
	2	Data 1
	3	Data 2
	4	Data 3
	5	Data 4
	6	Data 5
	7	Data 6
	8	Data 7
	9	Address 0
	10	Address 1
	11	Address 2
	12	Address 3
	13	R/W direction input
	14	ENB input
	15	RESET input
	16	Interruption output
	17	Serial input signal (CH0)
	18	Serial output signal (CH0)
	19	+ 5V
	20	GND

(11) XJ386: Power supply for stack

Connector No.	Terminal No.	Signal
XJ386 Power supply for stack	1	+ V 5
	2	+ V 5
	3	+ V 5
	4	+ V 5
	5	GND
	6	GND
	7	GND
	8	GND
	9	GND
	10	GND

3.4 Extended I/O BOARD (P-5500*)



Connector function in details

(1) XJ500 - XJ519 Input connector: PH connector (3P)

Connector No.	Terminal No.	Signal
XJ500 - XJ519 Input connector: PH connector (3P)	1	I24V: The voltage is consistently supplied.
	2	IN-SIG: Short circuit in pins 2 & 3; Input ON, Released in pins 2 & 3; Input OFF
	3	PGND

(2) XJ530 - XJ537 Output connector: PH connector (4P)

Connector No.	Terminal No.	Signal
XJ530 - XJ537 Output connector: PH connector (4P)	1	O24V: The voltage is shut off in an emergency stop.
	2	OUT-SIG: The terminal is directly connected to PGND in signal output mode.
	3	PGND
	4	Not used

(3) XJ538 - XJ553 Output connector: PH connector (2P)

Connector No.	Terminal No.	Signal
XJ538 - XJ553 Output connector: PH connector (2P)	1	O24V: The voltage is shut off in an emergency stop.
	2	OUT-SIG: The terminal is directly connected to PGND in signal output mode.

(4) XJ560: Connector for bus connection; PH connector, PCR Connector (34P)

They are connectors for connecting with other boards (P5530* for TSC-RS-120).

3.5 Control Panel Items (Maintenance Service) (Option)

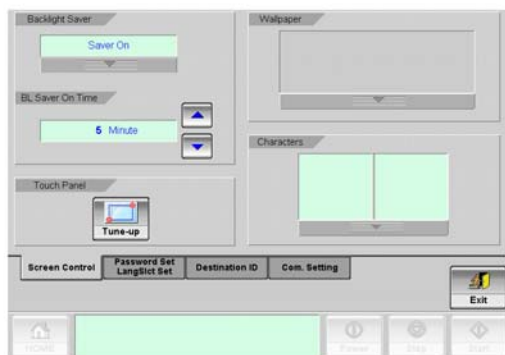


Fig. 3-1 [Main Menu Screen with Control Panel Opened]

3.5.1 Display Control Menu Screens

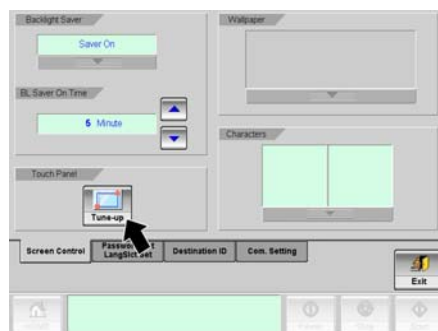
3.5.1.1 Touch Panel Coordinate Adjustment

TIP

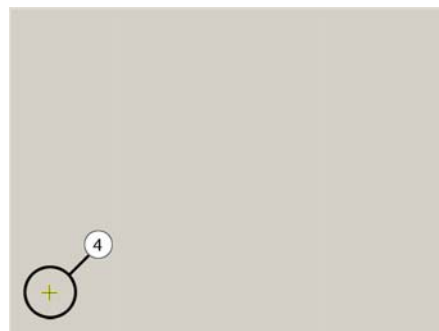
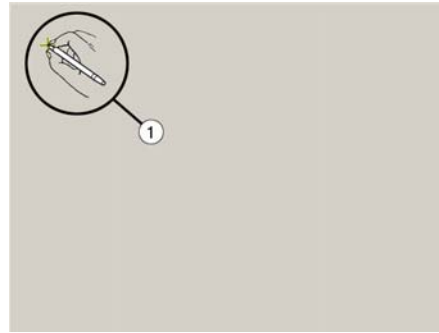
When the other page such as "password setting" is opened right before the operation, press the [Screen Control] menu.



1. Press the [Tune-up] key on the [Display Control menu] screen.
2. The [Confirmation Message] appears, and press the [Yes] key. The [Touchkit] screen appears.
3. Press the [Cal 4 Point] key.



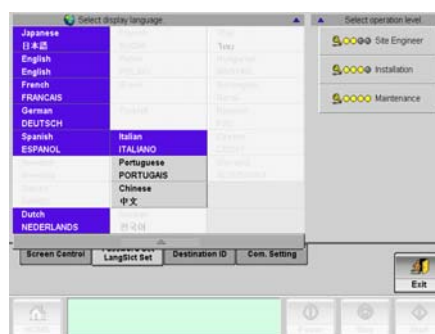
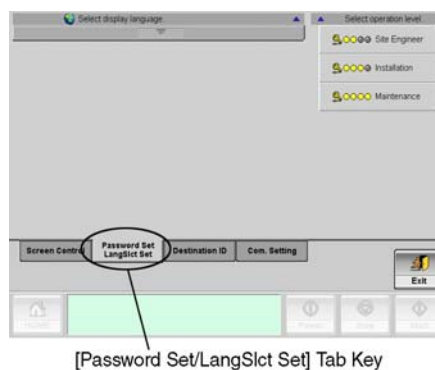
4. The following screens appear, and touch on the mark at each coordinate position with a ball point pen, etc.
5. Touch on the coordinate position mark displayed automatically in order. After 4 points are touched, the [Confirmation Message] appears automatically.



3.5.2 Password Set/Language Select Set Menu Screen

3.5.2.1 Language Select Setting

1. Press the [Password Set/LangSlct Set] tab key.
The Password Set/Language Select Set menu screen appears. Press the [Select display language] pop-up key.
2. Select a display language.



3.5.2.2 Password Setting

■ Press the key to change the password on the [Select operation level].

After entering password for change, press the [Return] key.

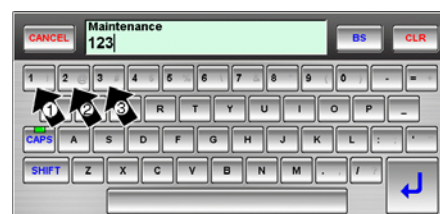
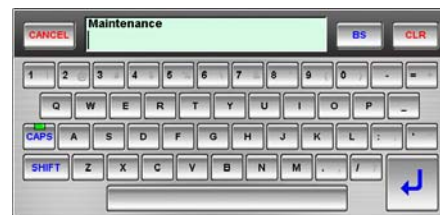
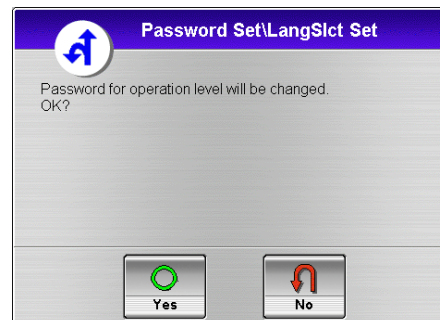
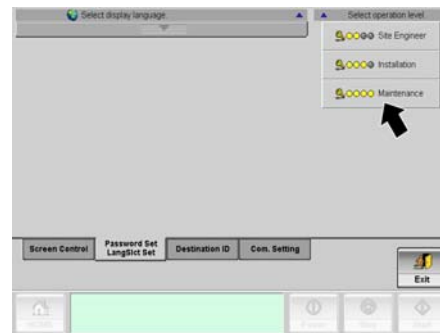


Fig. 3-2 [Example]

3.5.3 Destination ID Menu Screen

3.5.3.1 Destination ID

1. Press the [Destination ID] tab key. The Destination ID menu screen appears.



2. Press the [Destination ID] pop-up key, and select one from [Off, Printer, Card or E-mail].

TIP

Usually, select the [Printer].



3.5.3.2 Browser Setting

- Set the [Access Address] accordingly.

3.5.3.3 E-mail Setting

- Set each item of [Subject, TO 1...etc.] accordingly.

TIP

This equipment should be connected to the network before setting.

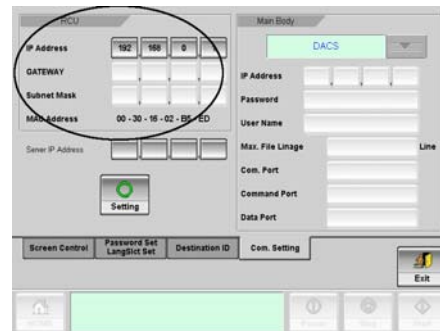


3.5.4 Communication Set Menu Screen

- Touch on the [Com. Setting] tab key. The [Communication Setting] menu screen appears.

3.5.4.1 [RCU] Communication Setting

- Set the [IP Address, GATEWAY, Subnet Mask] in RCU.



3.5.4.2 [Main Body] Communication Setting

- Set the [IP Address, Password, User Name, etc.] in CCM selected in the Main Body.



3.5.4.3 [Server IP Address] Setting

- Set the Server IP Address.

NOTE

Be sure not to make an incorrect change. Otherwise, it may cause the communication failure.

TIP

- The standard values are set at the factory; use them as they are.
- Change these values when interlocking with the computer after receiving the equipment.



4 EXPERT SETTINGS

4.1	Machine Parameter screen	4-1
4.1.1	Drive Unit Setting screen	4-2
4.1.2	PID Control Setting Screen	4-3
4.1.3	Motor Setting Screen	4-4
4.1.4	System Diagnosis screen	4-5
4.1.4.1	System Diagnosis screen 1 (GPIO)	4-6
4.1.4.2	System Diagnosis screen 3 (BL-MOTOR)	4-8
4.1.4.3	System Diagnosis screen 6 (OTHER)	4-9
4.1.5	Program Number Screen	4-10
4.1.6	Memory Inti. & Copy Screen	4-10

<MEMO>